

Targeted Therapy & Precision Oncology

Kinase Inhibitors, Antibody-Drug Conjugates, and Biomarker-Driven Treatment

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1. Rationale for Targeted Therapy

Targeted agents exploit molecular dependencies ('oncogene addiction') of tumour cells, achieving superior therapeutic indices compared with conventional cytotoxics. Selection requires validated companion diagnostics (CDx) that identify the actionable alteration in a patient's tumour. The FDA now approves tumour-agnostic indications (e.g., larotrectinib for NTRK-fusion positive solid tumours) where a single molecular marker predicts response regardless of histology.

Basket trials enroll patients based on molecular alteration rather than tumour site, enabling efficient evaluation of targeted agents across rare populations.

2. Small-Molecule Kinase Inhibitors (TKIs)

Target	Agent(s)	Generation	Key Indication	Resistance Mechanism
EGFR exon19del/L858R	Osimertinib	3rd	NSCLC 1st line	C797S; MET amp; HER2 amp
ALK	Alectinib, Lorlatinib	2nd / 3rd	ALK+ NSCLC	ALK G1202R compound mut.

BCR-ABL1	Imatinib / Ponatinib	1st / 3rd	CML, Ph+ ALL	T315I (ponatinib overcomes)
BRAF V600E	Dabrafenib + Trametinib	—	Melanoma, NSCLC, CRC	MAPK reactivation, RAS mut.
BRCA1/2	Olaparib, Niraparib	—	Ovarian, Breast, PDAC, PC	Reversion mutations, ABCB1
CDK4/6	Palbociclib, Ribociclib	—	ER+ HER2- Breast	RB loss, CDK6 amplification
FLT3	Midostaurin, Gilteritinib	—	FLT3-mut AML	F691L gatekeeper mutation
NTRK1/2/3	Larotrectinib, Entrectinib	—	NTRK fusion (any solid)	NTRK kinase domain mut.

3. Monoclonal Antibodies

Agent	Target	Mechanism	Key Indication
Trastuzumab	HER2	ADCC + HER2 signaling blockade	HER2+ Breast, Gastric
Cetuximab	EGFR	ADCC + receptor blockade	RAS-wt CRC, SCCHN
Bevacizumab	VEGF-A	Anti-angiogenesis	CRC, NSCLC, GBM, Ovarian
Rituximab	CD20	ADCC + CDC + apoptosis	B-cell lymphomas, CLL
Daratumumab	CD38	ADCC + immunomodulation	Multiple myeloma
Denosumab	RANK-L	Osteoclast inhibition	Bone metastases; GCTB

4. Antibody-Drug Conjugates (ADCs)

ADCs consist of a tumour-targeting antibody linked via a chemical linker to a cytotoxic payload. Key design parameters include target expression level, linker stability (cleavable vs. non-cleavable), payload potency (DAR — drug-to-antibody ratio, typically 2-8), and bystander effect. Three FDA-approved ADCs have transformed breast cancer management:

- Trastuzumab emtansine (T-DM1) — HER2 targeted; DM1 (maytansinoid) payload; HER2+ breast 2nd line
- Trastuzumab deruxtecan (T-DXd) — HER2 targeted; topoisomerase I inhibitor payload; high bystander effect; HER2+ and HER2-low breast/gastric
- Sacituzumab govitecan — TROP-2 targeted; SN-38 payload; TNBC and HR+ HER2- breast

5. Companion Diagnostics (CDx)

FDA-approved CDx assays are mandated for prescribing specific targeted therapies. Platforms include IHC (PD-L1, HER2, ER/PR), FISH/ISH (ALK, HER2, RET), PCR-based assays, and comprehensive genomic profiling (CGP) via NGS panels (e.g., FoundationOne CDx, MSK-IMPACT). Liquid biopsy (ctDNA) CDx platforms are approved for

EGFR mutation detection (cobas EGFR v2) and BRCA1/2 status (BRACAnalysis CDx in plasma).

CLINICAL PEARL: Tumour heterogeneity and clonal evolution mean that tissue biopsy at progression — or liquid biopsy ctDNA — is recommended to characterise resistance mechanisms before initiating subsequent-line therapy.